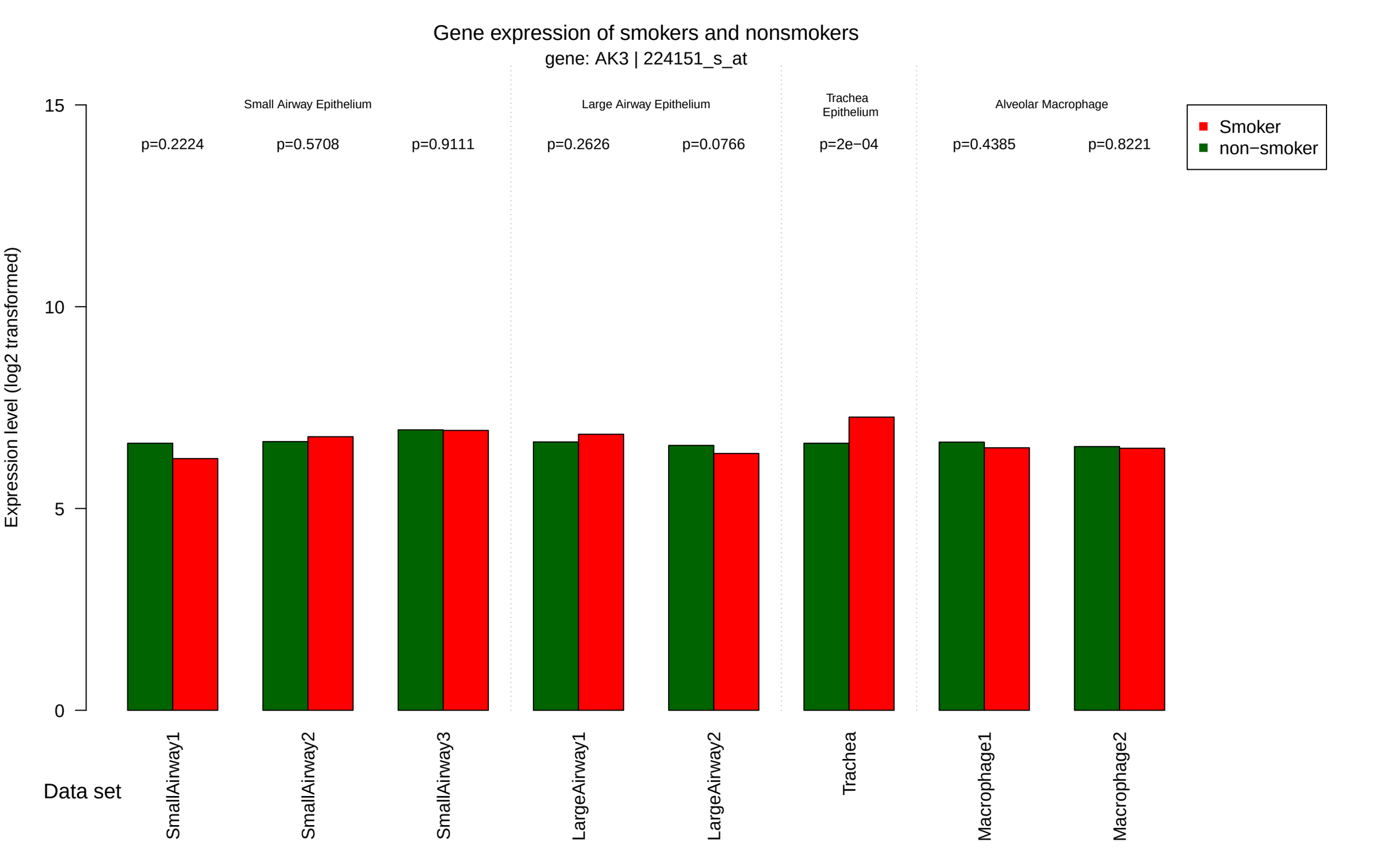
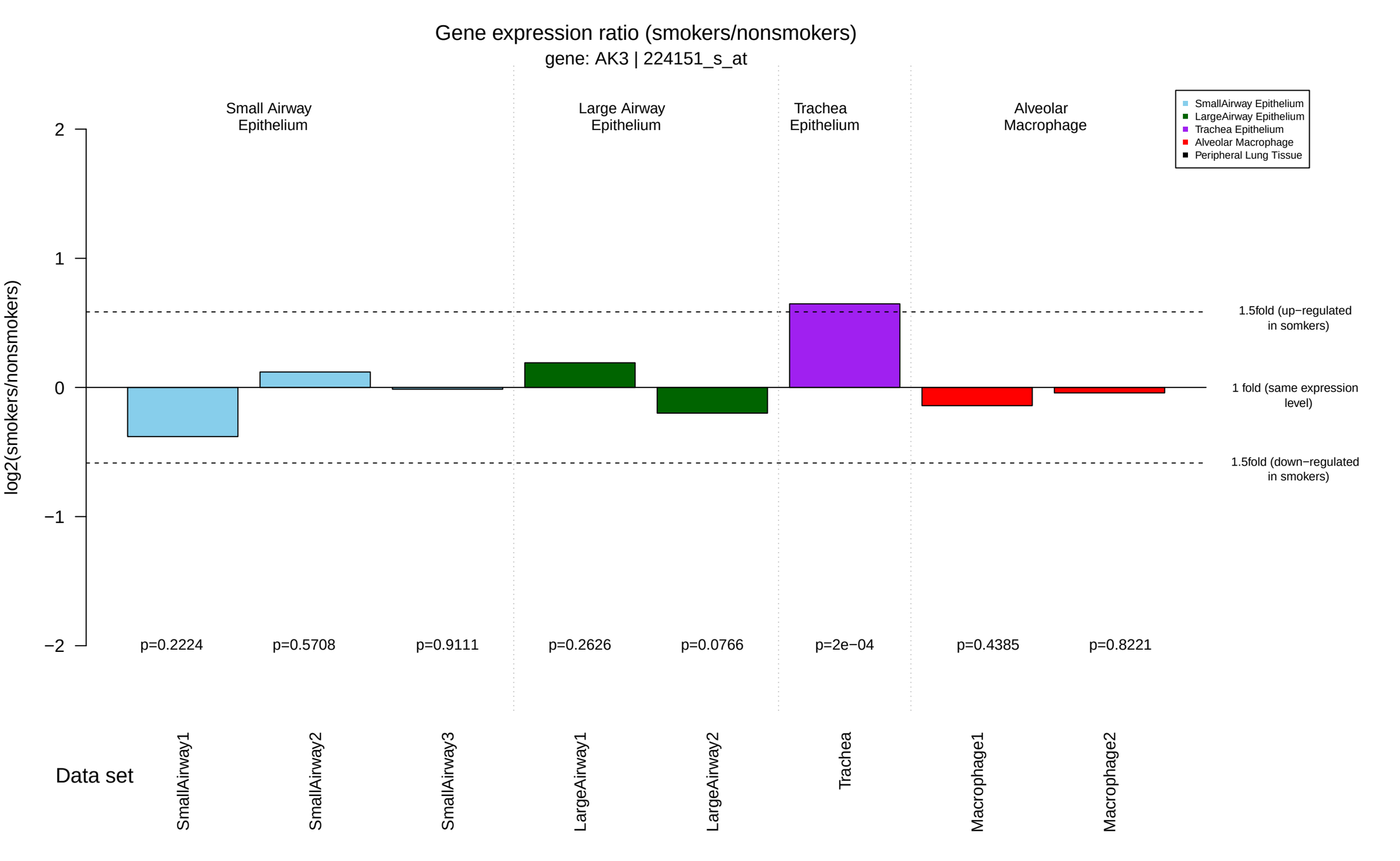
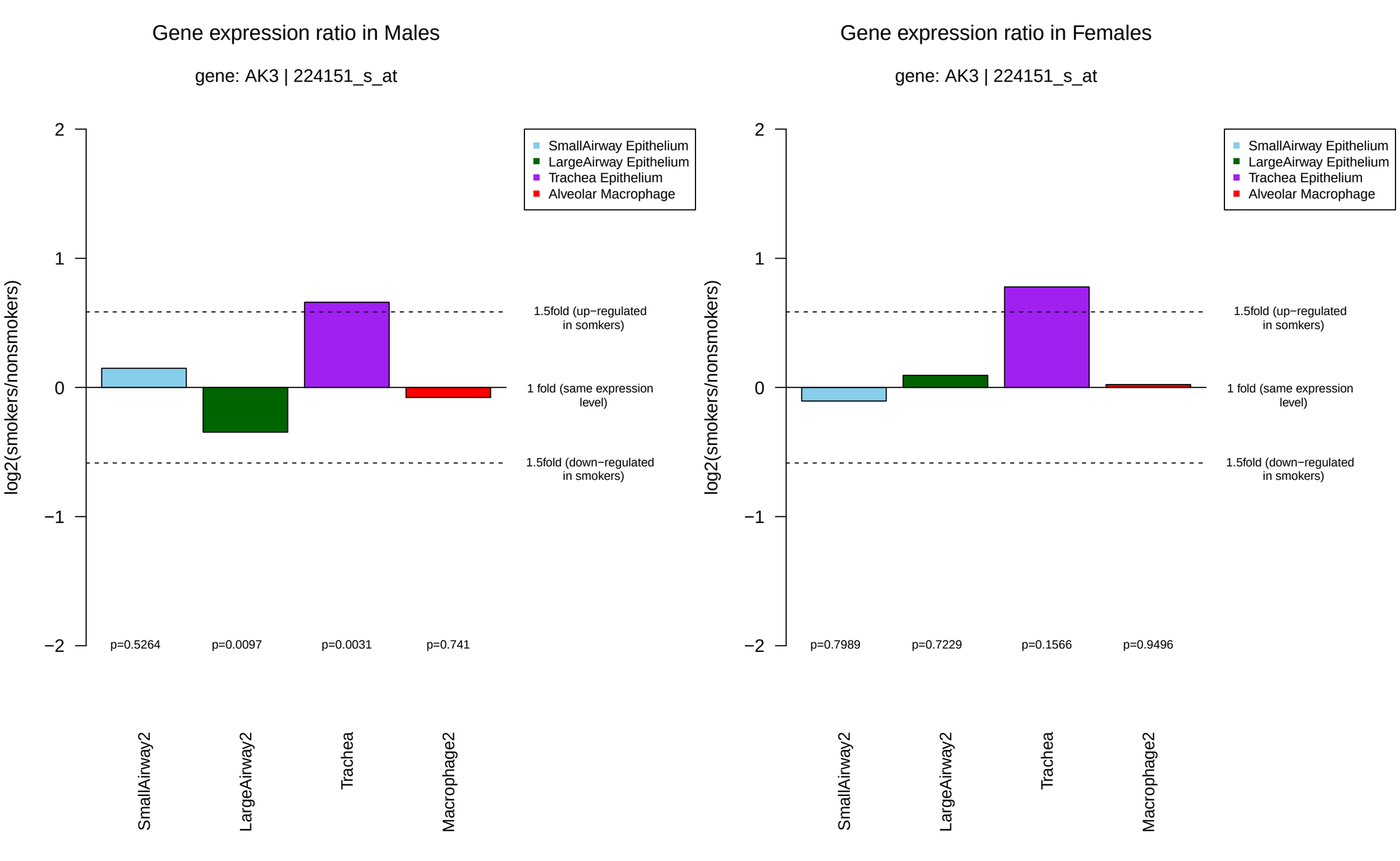
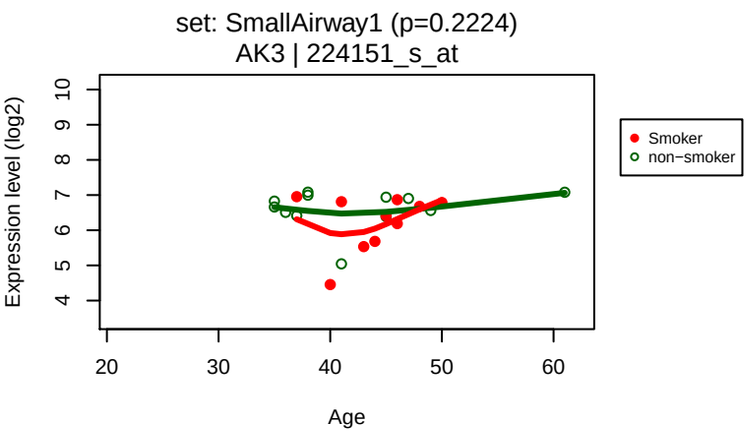
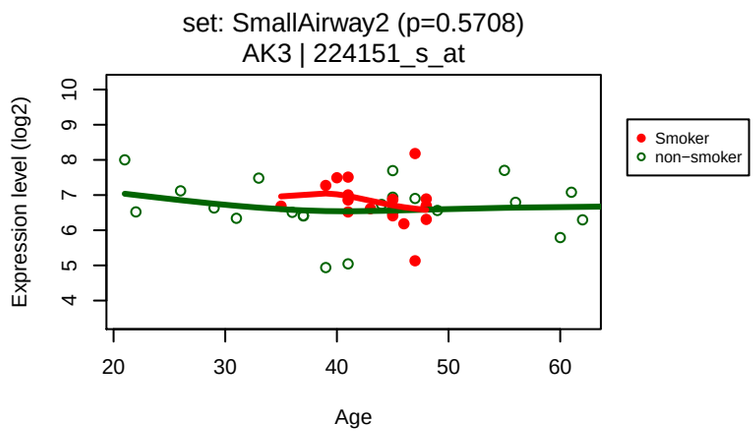
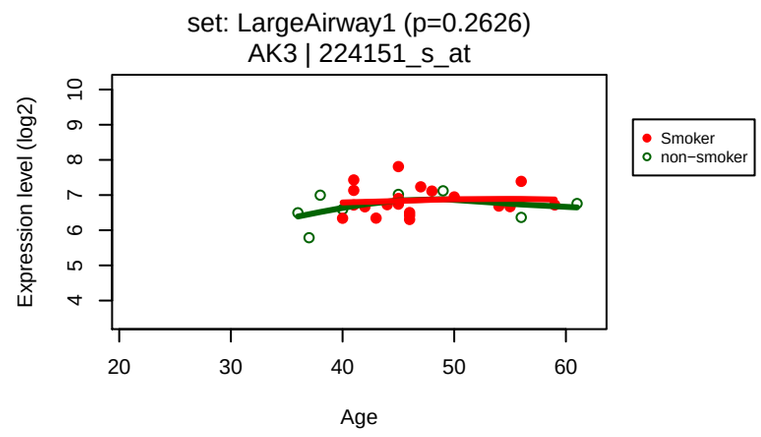

Gene expression of smokers and nonsmokers
gene: AK3 | 224151_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2224
p=0.5708
p=0.9111
p=0.2626
p=0.0766
p=2e−04
p=0.4385
p=0.8221
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

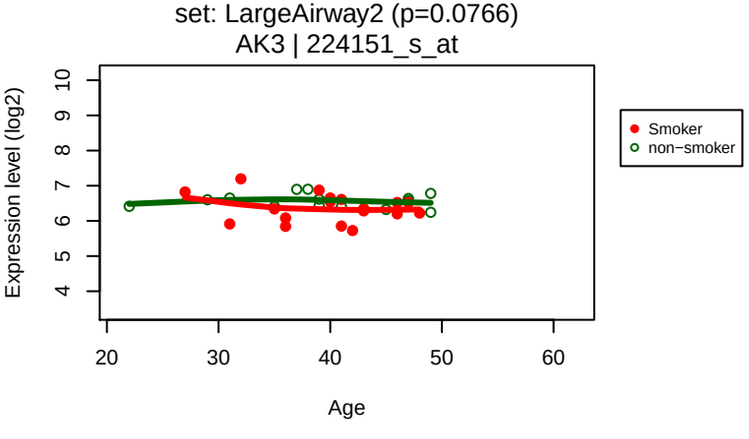

Gene expression ratio (smokers/nonsmokers)
gene: AK3 | 224151_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2224
p=0.5708
p=0.9111
p=0.2626
p=0.0766
p=2e−04
p=0.4385
p=0.8221
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

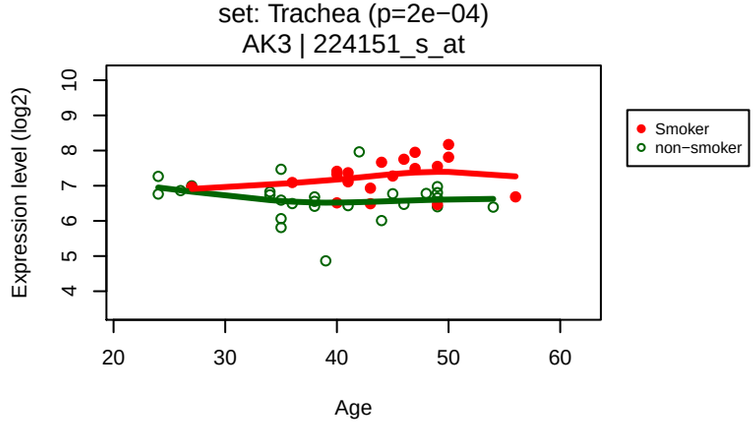

Gene expression ratio in Males
gene: AK3 | 224151_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5264
p=0.0097
p=0.0031
p=0.741
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AK3 | 224151_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7989
p=0.7229
p=0.1566
p=0.9496
SmallAirway2
LargeAirway2
Trachea
Macrophage2

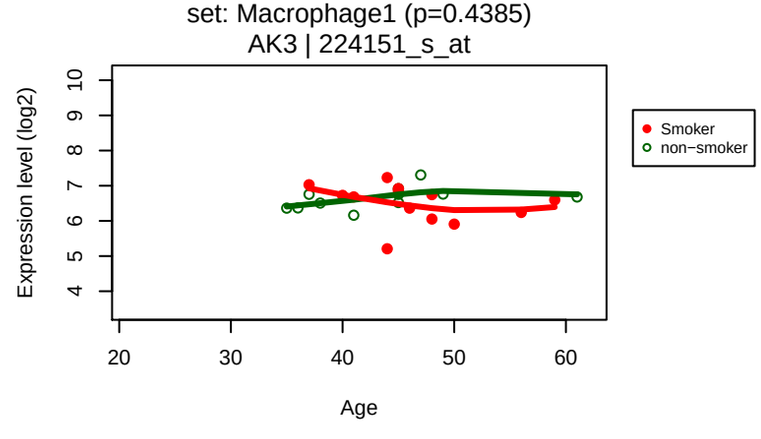

set: SmallAirway1 (p=0.2224)
AK3 | 224151_s_at
Expression level (log2)
Age
Smoker
non−smoker

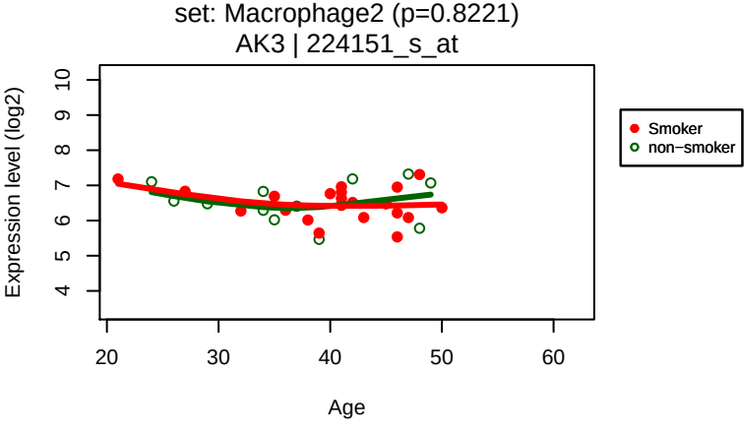

set: SmallAirway2 (p=0.5708)
AK3 | 224151_s_at
Expression level (log2)
Age
Smoker
non−smoker

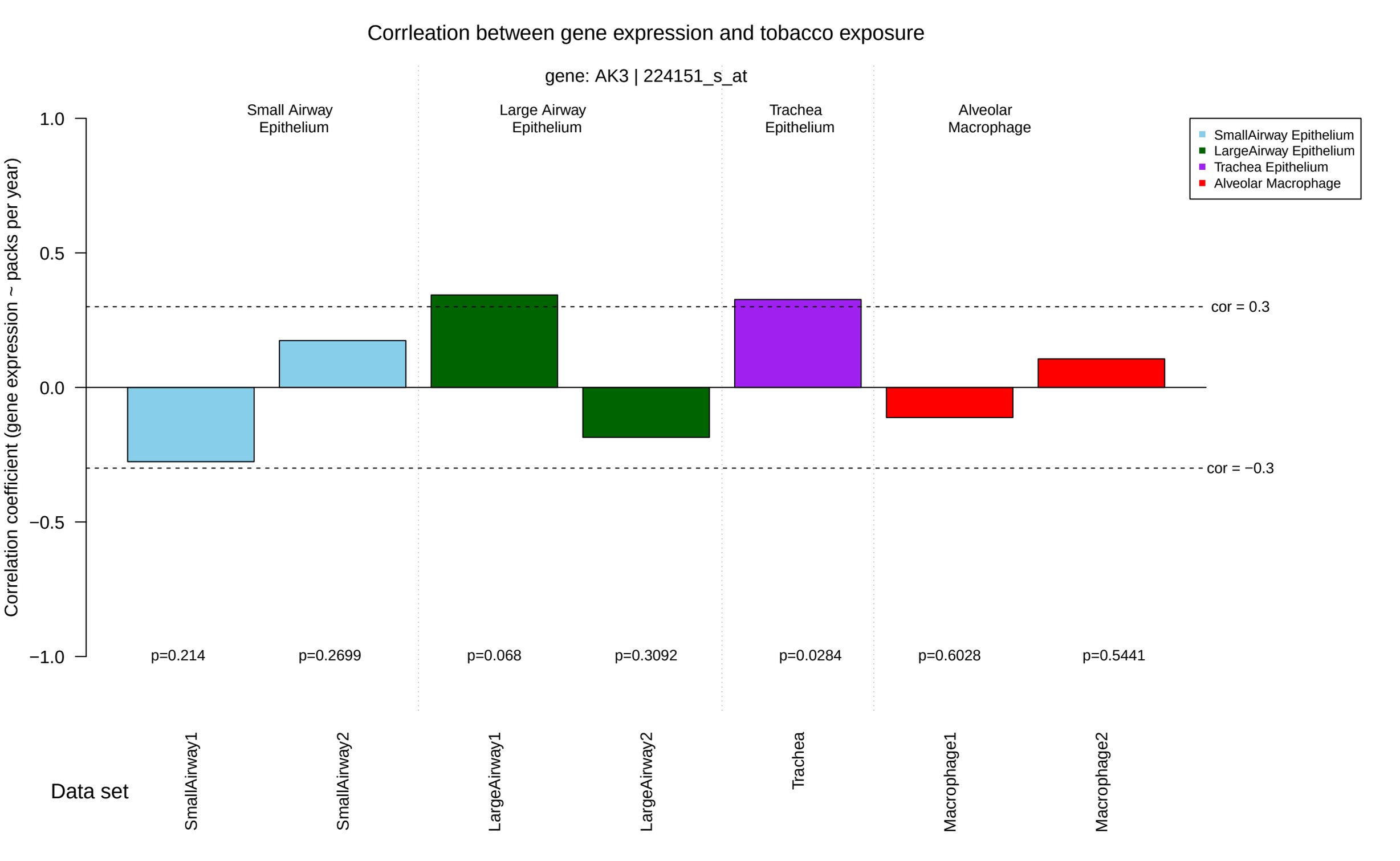

set: LargeAirway1 (p=0.2626)
AK3 | 224151_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0766)
AK3 | 224151_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
AK3 | 224151_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4385)
AK3 | 224151_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8221)
AK3 | 224151_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AK3 | 224151_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.214
p=0.2699
p=0.068
p=0.3092
p=0.0284
p=0.6028
p=0.5441
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2